



B.TECH. PROJECT WORK

Guidelines for Evaluation of Project Work Applicable to all Streams

1. Evaluation Scheme

The Project Work is intended to provide students with an opportunity to apply engineering knowledge, analytical ability, design thinking, problem-solving skills, research aptitude, innovation and professional practices to a well-defined engineering problem. The evaluation shall assess not merely the final outcome but also the systematic progression of the work, including identification of the problem, review of existing knowledge, formulation of objectives, development and execution of methodology, analysis and validation of results, documentation, presentation and technical defense.

The B. Tech. Project Work shall consist of Project Phase I and Project Phase II. Phase I shall have a common evaluation scheme for all students, irrespective of the track selected. Track-specific evaluation shall be introduced only in Phase II.

2. Project Tracks

The B. Tech. Project Work will be under any one of the following tracks,

- Track I – Industry Internship with Capstone Project
- Track II – Entrepreneurship Venture Track
- Track III – Higher Education Opportunity Track

Evaluation Principle

Project Phase I – Common Evaluation for Track I, Track II and Track III

Project Phase II – Separate Evaluation for Track I and Common Evaluation for Track II & III

3. Overall Evaluation Scheme

Project Phase	Applicability	Term Work (Internal)	POE (External)	Total
Project Phase I	Common – All Tracks	50	–	50
Project Phase II	Track I	50	50	100
Project Phase II	Track II & III	200	100	300

4. Evaluation Schedule

Phase	Applicability	Evaluation	TW	POE
Phase I	All Tracks	Presentation I	25	–
Phase I	All Tracks	Presentation II	25	–
Phase II	Track I	Presentation III	25	–
Phase II	Track I	Presentation IV	25	–
Phase II	Track I	Final POE	–	50
Phase II	Track II & III	Presentation III	100	–
Phase II	Track II & III	Presentation IV	100	–
Phase II	Track II & III	Final POE	–	100



5. General Evaluation and Progression Requirements

- [1] Each student/student group shall carry out the Project Work under the supervision of an approved Project Guide.
- [2] Project Phase I shall be common to all tracks and shall establish the project/problem foundation before detailed work is undertaken in Phase II.
- [3] Presentation I shall be completed before the student becomes eligible for Presentation II.
- [4] All observations, corrections and recommendations made during a presentation shall be documented and satisfactorily incorporated before the subsequent presentation.
- [5] A student/group whose performance is found unsatisfactory at any stage may be required to repeat the respective presentation.
- [6] Successful completion of Project Phase I shall be mandatory for progression to Project Phase II.
- [7] During Phase II, the student/group shall be evaluated according to:
 - a) Track I Evaluation Scheme, or
 - b) Common Track II & III Evaluation Scheme, as applicable.
- [8] The internal evaluation shall consider both the quality of work and continuous progress demonstrated by the student/group.
- [9] Only students who satisfactorily complete all prescribed internal evaluations shall be eligible for the Final Presentation and Oral Examination (POE).
- [10] The Final POE shall evaluate the overall quality and completeness of work, achievement of objectives, technical understanding, results/outcomes, documentation, presentation and ability to defend the work.

**B. TECH. PROJECT PHASE I****Presentation I**

(Maximum Marks: 25)

Track I Industry Problem Identification and Preliminary Proposal

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Name of Industry if applicable : _____

Sem. & Acad. Year : _____

Sr. No.	Evaluation Criterion	Expected / Relevant Contents	Marks	Marks Obtained
1	Industry Problem / Need and Relevance	Identification of a genuine industry/field problem, stakeholder need, engineering relevance and significance of the proposed work	5	
2	Study of Existing Industry Practice / System	Existing processes, systems, technologies, practices, limitations, constraints and industry requirements related to the identified problem	5	
3	Project Scope and Preliminary Objectives	Clearly defined scope, preliminary objectives and intended engineering deliverables aligned with the industrial problem	5	
4	Preliminary Technical Approach	Proposed design, analysis, development or implementation approach; preliminary tools, standards, software, equipment and resources	5	
5	Proposal Presentation and Technical Understanding	Organization, clarity, preparedness, technical understanding of the identified industrial problem and response to questions	5	
	TOTAL		25	

Review Comments / Corrections Required

- _____
- _____
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Evaluation Status

☐ Completed satisfactorily ☐ Corrections required**Supervisor****HoD/Chairperson**

**Presentation II**

(Maximum Marks: 25)

Track I Project Planning and Methodology Review

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Name of Industry if applicable : _____

Sem. & Acad. Year : _____

Sr. No.	Evaluation Criterion	Expected / Relevant Contents	Marks	Marks Obtained
1	Incorporation of Presentation-I Suggestions	Evidence of satisfactory incorporation of observations, corrections and recommendations from Presentation I	5	
2	Technical Literature / Codes / Existing-System Analysis	Relevant technical literature, standards, codes, specifications, industry practices and critical study of existing solutions	5	
3	Final Objectives and Engineering Deliverables	Specific, achievable and measurable project objectives with clearly identified technical deliverables	5	
4	Detailed Methodology, Feasibility and Work Plan	Design/development/implementation methodology, resources, industry facilities, technical feasibility, milestones and schedule	5	
5	Proposal Presentation and Technical Defence	Clarity of proposal, engineering justification, technical preparedness and response to questions	5	
	TOTAL		25	

Review Comments / Corrections Required

1. _____
2. _____
3. _____
4. _____
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Supervisor**HoD/Chairperson**



Phase-I Completion

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Name of Industry if applicable : _____

Sem. & Acad. Year : _____

Assessment	Maximum Marks	Marks Obtained	Status
Presentation I	25		
Presentation II	25		
Phase-I Term Work	50		

Supervisor

HoD/Chairperson

**B.TECH. PROJECT PHASE II****Presentation III**

Maximum Marks: 25

Track I - Industry Project Planning and Progress Review

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Name of Industry if applicable : _____

Sem. & Acad. Year : _____

Sr. No.	Evaluation Criterion	Expected / Relevant Contents	Marks	Marks Obtained
1	Understanding of Industry Problem and Requirements	Understanding of actual industry problem, stakeholder requirements, scope and constraints	5	
2	Study of Existing System and Technical Feasibility	Existing practices/solutions, limitations and feasibility of proposed solution	5	
3	Engineering Design / Development Methodology	Design approach, specifications, calculations, tools, standards and procedures	5	
4	Internship Engagement, Project Planning and Progress	Industry engagement, responsibilities, schedule, progress and maintenance of records	5	
5	Presentation and Technical Understanding	Clarity, technical preparedness and response to questions	5	
	TOTAL		25	

Review Comments / Corrections Required

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Evaluation Status☐ Completed Satisfactorily ☐ Corrections Required**Supervisor****HoD / Chairperson**

**Presentation IV**

Maximum Marks: 25

Track I - Implementation and Pre-Final Review

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Name of Industry if applicable : _____

Sem. & Acad. Year : _____

Sr. No.	Evaluation Criterion	Expected / Relevant Contents	Marks	Marks Obtained
1	Incorporation of Previous Review Suggestions	Satisfactory incorporation of prescribed corrections/modifications	5	
2	Design / Development / Field Implementation	Quality and progress of design, fabrication, development, modelling, construction or field implementation	5	
3	Testing, Analysis and Validation	Testing procedures, results, verification, validation and interpretation	5	
4	Achievement of Objectives and Completion Readiness	Progress against objectives, completion status and readiness for final evaluation	5	
5	Demonstration, Presentation and Technical Defence	Demonstration of work, communication and ability to answer technical questions	5	
	TOTAL		25	

Review Comments / Corrections Required

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Evaluation Status☐ Completed Satisfactorily ☐ Corrections Required**Supervisor****Industry Mentor (If applicable)****HoD / Chairperson**



Phase-II Completion

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Name of Industry if applicable : _____

Sem. & Acad. Year : _____

Assessment	Maximum Marks	Marks Obtained	Status
Presentation I	25		
Presentation II	25		
Phase-I Term Work	50		

Supervisor

HoD/Chairperson

**Final Presentation and Oral Examination**

Maximum POE: 50 Marks

Track I - Final Presentation and Oral Examination

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Name of Industry if applicable : _____

Sem. & Acad. Year : _____

Sr. No.	Evaluation Criterion	Expected / Relevant Contents	Marks	Marks Obtained
1	Completion and Industry Relevance	Completeness and relevance of solution to identified industry/field problem	5	
2	Technical Quality and Achievement of Objectives	Engineering correctness, technical depth and achievement of objectives	10	
3	Implementation, Demonstration and Validation	Quality of implementation, working demonstration, testing and validation	10	
4	Industry Learning and Professional Application	Evidence of industry learning and application of professional engineering practices	5	
5	Project Report and Technical Documentation	Technical content, drawings, calculations, figures/tables, references and documentation	5	
6	Results, Conclusions and Future Scope	Interpretation, conclusions, limitations, recommendations and future scope	5	
7	Presentation and Oral Technical Defence	Communication, subject knowledge, technical justification and response to questions	10	
	TOTAL		50	

Supervisor**External Examiner**

**B. TECH. PROJECT PHASE I****Presentation I**

(Maximum Marks: 25)

Track II Entrepreneurial Opportunity and Venture Proposal Review

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Semester : _____

Academic Year : _____

Sr. No.	Evaluation Criterion	Expected / Relevant Contents	Marks	Marks Obtained
1	Problem / Customer Need Identification	Identification of a genuine customer/user problem, unmet need or business opportunity supported by preliminary observations	5	
2	Market and Opportunity Relevance	Target customer segment, potential market, preliminary demand, industry relevance and significance of the opportunity	5	
3	Proposed Solution and Value Proposition	Proposed product/service/solution, customer value, innovation and differentiation from available alternatives	5	
4	Preliminary Technical and Business Feasibility	Preliminary technical requirements, resources, infrastructure, cost considerations and feasibility of the proposed venture	5	
5	Venture Proposal / Pitch and Q&A	Clarity of business idea, entrepreneurial understanding, communication and response to questions	5	
	TOTAL		25	

Review Comments / Corrections Required

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Evaluation Status☐ Completed satisfactorily ☐ Corrections required**Supervisor****HoD/Chairperson**

**Presentation II**

(Maximum Marks: 25)

Track II Market Validation and Venture Planning Review

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Semester : _____

Academic Year : _____

Sr. No.	Evaluation Criterion	Expected / Relevant Contents	Marks	Marks Obtained
1	Incorporation of Presentation-I Suggestions	Refinement of identified problem, proposed solution, value proposition and venture concept based on review comments	5	
2	Customer / Market / Competitor Study	Customer/stakeholder interaction, market evidence, competitor analysis, alternatives and preliminary validation of demand	5	
3	Business Model and Value Proposition	Customer segments, channels, key activities/resources/partners, cost/revenue logic and refined value proposition	5	
4	Feasibility, MVP and Venture Work Plan	MVP/prototype/service development plan, technical and commercial feasibility, resources, risks, milestones and timeline	5	
5	Entrepreneurship Seminar / Pitch and Defence	Quality of entrepreneurial pitch, justification of assumptions, business understanding and response to questions	5	
	TOTAL		25	

Review Comments / Corrections Required

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Supervisor**HoD/Chairperson**



Sanjay Ghodawat University Kolhapur

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Phase-I Completion

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Semester : _____

Academic Year : _____

Assessment	Maximum Marks	Marks Obtained	Status
Presentation I	25		
Presentation II	25		
Phase-I Term Work	50		

Supervisor

HoD/Chairperson

**B.TECH. PROJECT PHASE II****Presentation III**

Maximum Marks: 25

Track II - Venture Development and Validation Review

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Semester : _____

Academic Year : _____

Sr. No.	Evaluation Criterion	Expected / Relevant Contents	Marks	Marks Obtained
1	Incorporation of Phase-I Recommendations	Refinement of opportunity, solution and venture plan	10	
2	Business Opportunity, Customer Need and Value Proposition	Clarity of problem, opportunity, target customer and value proposition	15	
3	Customer / Market / Competitor Validation	Customer interaction, market evidence and competitor validation	15	
4	MVP / Prototype / Product-Service Development	Progress and quality of MVP, prototype, product or service	20	
5	Technical and Operational Feasibility	Technology, infrastructure, resources, operations and implementation feasibility	15	
6	Preliminary Business Model and Revenue Strategy	Business model, revenue mechanism, channels, resources and preliminary pricing	10	
7	Venture Progress, Risks and Problem Solving	Progress, risks, corrective action and entrepreneurial problem solving	5	
8	Documentation, Pitch and Entrepreneurial Defence	Documentation, pitch quality and response to questions	10	
	TOTAL		100	

Review Comments / Corrections Required

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Evaluation Status☐ Completed Satisfactorily ☐ Corrections Required**Supervisor****HoD / Chairperson**

**Presentation IV**

Maximum Marks: 100

Track II - Techno-Economic and Venture Readiness Review

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Semester : _____

Academic Year : _____

Sr. No.	Evaluation Criterion	Expected / Relevant Contents	Marks	Marks Obtained
1	Incorporation of Presentation-III Corrections	Incorporation of previous recommendations and validation feedback	10	
2	Market Validation and Final Value Proposition	Strength of market/customer evidence and final value proposition	15	
3	MVP / Prototype / Service Readiness	Functionality, testing, feedback and maturity of proposed solution	15	
4	Business Model and Operational Plan	Final business model, operations, resources, partners and delivery mechanism	15	
5	Financial and Economic Feasibility	Cost, pricing, working capital, revenue, break-even/profitability and funding	20	
6	Go-to-Market, Scalability and Growth Strategy	Market entry, customer acquisition, growth strategy and scalability	10	
7	Risk, Regulatory and Sustainability Considerations	Risks, mitigation, regulatory requirements, ethics and sustainability	5	
8	Venture Report / Business Plan and Pitch	Completeness and professional quality of venture documentation and pitch	10	
	TOTAL		100	

Review Comments / Corrections Required

1. _____
2. _____
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Evaluation Status☐ Completed Satisfactorily ☐ Corrections Required**Supervisor****HoD / Chairperson**

**Track-II Internal Assessment**

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Semester : _____

Academic Year : _____

Assessment	Maximum Marks	Marks Obtained	Status
Presentation III	100		
Presentation IV	100		
Phase-II Term Work	200		

Supervisor**HoD / Chairperson**

**Track II - Final Venture Presentation and Oral Examination**

Maximum POE: 100 Marks

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Semester : _____

Academic Year : _____

Sr. No.	Evaluation Criterion	Expected / Relevant Contents	Marks	Marks Obtained
1	Business Idea, Innovation and Market Relevance	Need, innovation, differentiation and relevance to target market	10	
2	Customer and Market Validation	Quality and adequacy of customer and market evidence	15	
3	Technical Feasibility and MVP / Prototype	Technical soundness, functionality and implementation feasibility	15	
4	Business Model and Commercial Viability	Business model, operations, revenue mechanism and commercial viability	15	
5	Financial Feasibility and Funding Strategy	Costing, pricing, projections, break-even/profitability and funding	15	
6	Implementation / Go-to-Market / Scalability	Commercialization, market-entry strategy and growth potential	10	
7	Venture Report / Business Plan / DPR	Completeness, coherence and professional quality	5	
8	Final Pitch	Structure, clarity, communication and persuasiveness	5	
9	Entrepreneurial Oral Defence	Entrepreneurial understanding, decision-making and response to expert questions	10	
	TOTAL		100	

Supervisor**External Examiner**

**Presentation I**

(Maximum Marks: 25)

Track III Research Problem and Preliminary Research Proposal Review

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Semester : _____

Academic Year : _____

Sr. No.	Evaluation Criterion	Expected / Relevant Contents	Marks	Marks Obtained
1	Research Problem, Motivation and Relevance	Clearly identified research problem, significance, scope and technical/academic/industrial/societal relevance	5	
2	Preliminary Literature Review	Appropriate and recent scholarly literature, existing knowledge, methodologies, findings and limitations	5	
3	Research Gap Identification	Critical understanding of existing research and preliminary identification of an unresolved research/technical gap	5	
4	Research Objectives and Expected Contribution	Preliminary research objectives/questions and anticipated contribution, novelty or expected research outcome	5	
5	Research Proposal Presentation and Q&A	Organization, clarity, research understanding, preparedness and response to questions	5	
	TOTAL		25	

Review Comments / Corrections Required

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Evaluation Status☐ Completed satisfactorily ☐ Corrections required**Supervisor****HoD/Chairperson**

**Presentation II**

(Maximum Marks: 25)

Track III Research Methodology and Proposal Finalization Review

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Semester : _____

Academic Year : _____

Sr. No.	Evaluation Criterion	Expected / Relevant Contents	Marks	Marks Obtained
1	Incorporation of Presentation-I Suggestions	Refinement of research problem, literature review, gap and objectives based on previous recommendations	5	
2	Literature Synthesis and Final Research Gap	Critical synthesis of relevant scholarly work and clear justification of the research gap	5	
3	Final Objectives / Research Questions / Hypothesis	Specific and measurable objectives, questions/hypotheses aligned with the identified research gap	5	
4	Research Methodology, Feasibility and Work Plan	Experimental, numerical, computational, analytical, simulation or survey methodology; resources, data requirements and schedule	5	
5	Research Seminar and Technical Defence	Quality of research presentation, justification of methodology and response to technical/research questions	5	
	TOTAL		25	

Review Comments / Corrections Required

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Supervisor**HoD/Chairperson**

**Phase-I Completion**

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Semester : _____

Academic Year : _____

Assessment	Maximum Marks	Marks Obtained	Status
Presentation I	25		
Presentation II	25		
Phase-I Term Work	50		

Supervisor**HoD/Chairperson**

**Presentation III**

(Maximum Marks: 100)

Track III - Research Execution and Progress Review

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Semester : _____

Academic Year : _____

Sr. No.	Evaluation Criterion	Expected / Relevant Contents	Marks	Marks Obtained
1	Incorporation of Phase-I Recommendations	Satisfactory incorporation of proposal-stage recommendations	10	
2	Refinement of Literature and Research Framework	Updated literature, refined gap and research framework	10	
3	Implementation of Research Methodology	Systematic execution of approved research methodology	20	
4	Data Collection / Experimental / Numerical / Computational Work	Quality, adequacy and accuracy of research execution and data generation	20	
5	Preliminary Analysis and Interpretation	Preliminary analysis, calculations, trends and technical interpretation	15	
6	Validation / Verification Approach	Validation against theory, standards, experiments, literature or benchmark results	10	
7	Research Progress and Problem-Solving Ability	Progress against plan and handling of technical/research difficulties	5	
8	Documentation, Presentation and Research Defence	Research records, presentation, understanding and response to questions	10	
	TOTAL		100	

Review Comments / Corrections Required

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Evaluation Status☐ Completed Satisfactorily ☐ Corrections Required**Supervisor****HoD / Chairperson**

**Presentation IV**

(Maximum Marks: 100)

Track III - Final Research Progress Review

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Semester : _____

Academic Year : _____

Sr. No.	Evaluation Criterion	Expected / Relevant Contents	Marks	Marks Obtained
1	Incorporation of Presentation-III Corrections	Complete incorporation of previous observations and recommendations	10	
2	Completeness and Technical Quality of Research	Extent, technical rigor, accuracy and overall quality of completed work	15	
3	Data Analysis, Validation and Interpretation	Appropriate analysis, verification, validation and interpretation	20	
4	Results and Comparison with Literature / Standards	Quality of findings and critical comparison with published work/standards	15	
5	Achievement of Research Objectives	Extent to which stated research objectives/questions have been addressed	10	
6	Research Contribution / Novelty / Significance	Originality, contribution and significance of research outcomes	10	
7	Research Report / Manuscript Preparation	Academic writing, report quality, references and research-output progress	10	
8	Presentation and Research Defence	Research understanding, technical justification and response to questions	10	
	TOTAL		100	

Review Comments / Corrections Required

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Supervisor**HoD / Chairperson**

**Track-III Internal Assessment**

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Semester : _____

Academic Year : _____

Assessment	Maximum Marks	Marks Obtained	Status
Presentation III	100		
Presentation IV	100		
Phase-II Term Work	200		

Supervisor**HoD / Chairperson**

**Track III – Final Research Presentation and Oral Examination** Maximum POE: 100 Marks

Student Name : _____

PRN / Roll No. : _____

Program : _____

Department : _____

Supervisor Name : _____

Project Title : _____

Semester : _____

Academic Year : _____

Sr. No.	Evaluation Criterion	Expected / Relevant Contents	Marks	Marks Obtained
1	Quality, Relevance and Originality of Research	Significance, originality, relevance, technical depth and research quality	15	
2	Literature Foundation and Research Gap	Critical synthesis of literature and justification of research gap	10	
3	Research Methodology and Technical Rigor	Appropriateness, correctness and systematic execution of methodology	15	
4	Data Analysis, Verification and Validation	Accuracy of analysis, verification, validation and reliability of findings	15	
5	Results, Discussion and Achievement of Objectives	Findings, interpretation, comparison and achievement of objectives	15	
6	Research Contribution and Future Potential	Novelty, contribution, limitations, future research and publication/patent potential	10	
7	Research Report / Manuscript Quality	Academic writing, technical content, figures/tables and references	10	
8	Presentation	Structure, clarity and communication of research work	5	
9	Oral Examination / Research Defence	Subject knowledge, research understanding and response to examiner questions	5	
	TOTAL		100	

Supervisor**External Examiner**